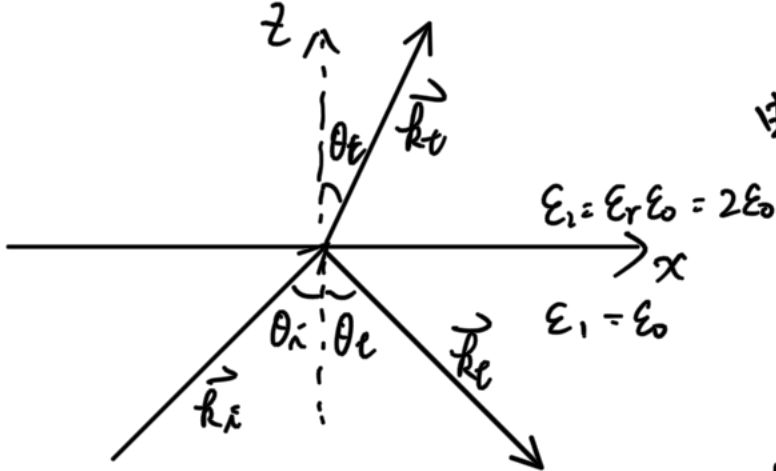


1. 一平面电磁波以 $\theta=45^\circ$ 从真空入射到 $\epsilon_r=2$ 的介质, 电场强度垂直于入射面。
利用折射率定律和边值关系, 求反射系数和折射系数。



由邊值關係知

$$\begin{cases} E_i + E_r = E_t \\ H_i \cos \theta_i - H_r \cos \theta_r = H_t \cos \theta_t \\ \theta_i = \theta_r, \quad H = \sqrt{\frac{\epsilon}{\mu}} E \end{cases}$$

根据菲涅爾公式知

$$\frac{E_r}{E_i} = - \frac{\sqrt{\epsilon_2} \cos \theta_t - \sqrt{\epsilon_1} \cos \theta_i}{\sqrt{\epsilon_2} \cos \theta_t + \sqrt{\epsilon_1} \cos \theta_i} = - \frac{n_{21} \cos \theta_t - \cos \theta_i}{n_{21} \cos \theta_t + \cos \theta_i} = - \frac{\sqrt{n_{21}^2 - \sin^2 \theta_i} - \cos \theta_i}{\sqrt{n_{21}^2 - \sin^2 \theta_i} + \cos \theta_i} = - \frac{\sqrt{1.5} - \sqrt{0.5}}{\sqrt{1.5} + \sqrt{0.5}} \quad \left(n_{21} = \frac{\sqrt{\epsilon_2}}{\sqrt{\epsilon_1}} = \sqrt{\epsilon_r} = \sqrt{2} \right) = -(2-\sqrt{3})$$

$$\frac{E_t}{E_i} = \frac{2\sqrt{\epsilon_1} \cos \theta_i}{\sqrt{\epsilon_1} \cos \theta_i + \sqrt{\epsilon_2} \cos \theta_t} = \frac{2 \cos \theta_i}{\cos \theta_i + \sqrt{n_{21}^2 - \sin^2 \theta_i}} = \frac{\sqrt{2}}{\sqrt{1.5} + \sqrt{0.5}} = \sqrt{3}-1$$

$$\text{反射系数} \quad R = \frac{\sqrt{\epsilon_i/\mu_i}}{\sqrt{\epsilon_r/\mu_r}} \left(\frac{E_r}{E_i} \right)^2 = \left(\frac{E_r}{E_i} \right)^2 = (2-\sqrt{3})^2 = 7-4\sqrt{3} \approx 0.0718$$

$$\text{折射系数} \quad T = \frac{\sqrt{\epsilon_r/\mu_r}}{\sqrt{\epsilon_i/\mu_i}} \left(\frac{E_t}{E_i} \right)^2 = n_{21} \left(\frac{E_t}{E_i} \right)^2 = \sqrt{2} \left(\sqrt{3}-1 \right)^2 = \sqrt{2} \times (4-2\sqrt{3}) = 4\sqrt{2}-2\sqrt{6} \approx 0.7579$$

2. 有一个可见平面光波由水入射到空气, 入射角为 60° 。证明这时将会发生全反射, 并求折射波（消逝场）沿表面传播的相速度和透入空气的深度。设该波在空气中的波长为 $\lambda_0=628\text{nm}$, 水的折射率为 $n=1.33$ 。

$$\text{此光波在空氣中的波文為} \quad k_t = \frac{2\pi}{\lambda_0} = \frac{2\pi}{628\text{nm}} \approx 0.01 \text{ nm}^{-1}$$

$$\text{在水中的波文是} \quad k_i = \frac{2\pi f}{v} = \frac{2\pi}{\lambda_0} n = n k_t = 0.0133 \text{ nm}^{-1}$$

$$\text{全反射角} \quad \theta_c \text{ 满足 } \sin \theta_c = \frac{1}{n} = \frac{1}{1.33} < \frac{1}{1.25} = 0.8$$

$$\text{而 } \sin \theta_i = \sin 60^\circ = \frac{\sqrt{3}}{2} > \frac{\sqrt{3}b}{2} = 0.8 > \frac{1}{1.33} = \sin \theta_c \Leftrightarrow \theta_i > \theta_c \Leftrightarrow \text{出現全反射}$$

$$\text{折射波沿表面傳播的相速度為} \quad v_s = \frac{\omega}{k_{tx}} = \frac{\omega}{k_{ix}} = \frac{\omega}{k_i \sin \theta_i} = \frac{\omega}{n k_t \sin \theta_i} = \frac{2\pi f}{n k_t \sin \theta_i} = \frac{c}{n \sin \theta_i} = \frac{3.0 \times 10^8}{1.33 \times \sin 60^\circ} \approx 2.605 \times 10^8 \text{ m/s}$$

$$\text{折射波振幅衰减系数} \quad \kappa = k_i \sqrt{\sin^2 \theta_i - \left(\frac{1}{n} \right)^2} = k_t \sqrt{(n \sin \theta_i)^2 - 1} = 5.716 \times 10^{-3} \text{ nm}^{-1}$$

$$\text{透入空氣深度} \quad \kappa^{-1} = 174.9 \text{ nm}$$

3. 入射光从介质 1 入射到介质 2 ($n_1 > n_2$), 当入射角大于全反射角时, 证明折射波沿线方向平均能流密度为 0。

$$\begin{aligned} \vec{S}_t &= \frac{1}{2} (\vec{E}_t + \vec{E}_t^*) \times \frac{1}{2} (\vec{H}_t + \vec{H}_t^*) \\ &= \frac{1}{4} (\vec{E}_t \times \vec{H}_t + \vec{E}_t^* \times \vec{H}_t^* + \vec{E}_t \times \vec{H}_t^* + \vec{E}_t^* \times \vec{H}_t) \\ &= \frac{1}{2} \text{Re} \{ \vec{E}_t \times \vec{H}_t \} + \frac{1}{2} \text{Re} \{ \vec{E}_t^* \times \vec{H}_t \} \\ &= \frac{1}{2} \text{Re} \{ \vec{E}_t(\vec{r}) \times \vec{H}_t(\vec{r}) e^{-i \cdot 2\omega t} \} + \frac{1}{2} \text{Re} \{ \vec{E}_t^*(\vec{r}) \times \vec{H}_t(\vec{r}) \} \end{aligned}$$

$$\langle \vec{S}_t \rangle = \frac{1}{2} \text{Re} \{ \vec{E}_t^* \times \vec{H}_t \}$$

$$\vec{E}_t = a (i \kappa \hat{x} + b \hat{y} - k_z \hat{z}) e^{-\kappa z} e^{-i k_x x}$$

$$\begin{aligned} \vec{E}_t^* \times \vec{H}_t &= \vec{E}_t^* \times \left(\frac{1}{\omega \mu} \vec{k} \times \vec{E}_t \right) = \frac{1}{\omega \mu} \left[|\vec{E}_t|^2 \vec{k} - (\vec{E}_t^* \cdot \vec{k}) \vec{E}_t \right] \\ &= \frac{|a|^2}{\omega \mu} \left[(\kappa^2 + b^2 + k_z^2) e^{-2\kappa z} (k_x \hat{x} + i \kappa \hat{z}) - (-i \cdot 2 k_z \kappa) e^{-2\kappa z} (i \kappa \hat{x} + b \hat{y} - k_z \hat{z}) \right] \end{aligned}$$

$$\Rightarrow \text{Re} \left\{ (\vec{E}_t^* \times \vec{H}_t) \right\} \cdot \hat{z} = 0$$

$$\text{平均能流密度在法向 (z 軸) 上為} \quad \langle \vec{S}_t \rangle_z = 0$$

4. 频率为 ω 的平面电磁波从真空垂直入射到厚度为 d 、折射率为 n 的介质膜, 计算反射系数 R , 并求出无反射（这种介质膜称为增透膜）的条件。

$$\begin{array}{c} \begin{array}{c} \uparrow z \\ \text{E}_{t2} \\ \downarrow d \\ \text{E}_{i1} = \text{E}_{i2} \quad \downarrow \quad \text{E}_{r2} \\ \uparrow \text{E}_{r1} = \text{E}_{t3} \end{array} \quad \begin{array}{c} \text{I} \\ \text{II} \\ \text{III} \end{array} \\ \text{E}_{i1} \quad \text{E}_{r1} = \text{E}_{t3} \end{array} \quad \begin{array}{c} \vec{E}_{i2} = (\text{E}_{i20} \hat{y}) e^{i[k_n(z-d) - \omega t]} \\ \vec{E}_{r2} = (\text{E}_{r20} \hat{y}) e^{i[-k_n(z-d) - \omega t]} \\ \vec{E}_{i1} = (\text{E}_{i10} \hat{y}) e^{i(k_0 z - \omega t)} \\ \vec{E}_{r1} = (\text{E}_{r10} \hat{y}) e^{i(-k_0 z - \omega t)} \end{array}$$

在 $z=d$ 處, 利用菲涅爾關係知 N 波满足

$$\frac{E_{r20}}{E_{i20}} = \frac{1-n^{-1}}{1+n^{-1}}$$

在 $z=0$ 處, 由邊值關係知

$$\begin{cases} E_{i10} + E_{r10} = E_{i20} e^{-i k_1 d} + E_{r20} e^{i k_1 d} = E_{i20} e^{-i k_1 d} + E_{r20} e^{i k_1 d} \\ H_{i10} - H_{r10} = H_{i20} e^{-i k_1 d} - H_{r20} e^{i k_1 d} \\ H = \sqrt{\frac{\epsilon}{\mu}} E \end{cases} \Rightarrow \begin{cases} E_{i10} + E_{r10} = E_{i20} \left(e^{-i k_1 d} + \frac{n-1}{n+1} e^{i k_1 d} \right) \\ E_{i10} - E_{r10} = n E_{i20} \left(e^{-i k_1 d} - \frac{n-1}{n+1} e^{i k_1 d} \right) \end{cases}$$

$$\Leftrightarrow \frac{E_{r10}}{E_{i10}} = \frac{n \left(e^{-i k_1 d} - \frac{n-1}{n+1} e^{i k_1 d} \right) - \left(e^{-i k_1 d} + \frac{n-1}{n+1} e^{i k_1 d} \right)}{n \left(e^{-i k_1 d} - \frac{n-1}{n+1} e^{i k_1 d} \right) + \left(e^{-i k_1 d} + \frac{n-1}{n+1} e^{i k_1 d} \right)} = \frac{(n-1) \left(e^{-i k_1 d} - e^{i k_1 d} \right)}{(n+1) e^{-i k_1 d} - \frac{(n-1)^2}{n+1} e^{i k_1 d}}$$

$$= \frac{-2i (n^2-1) \sin(k_1 d)}{4n \cos(k_1 d) - 2(n^2+1) i \sin(k_1 d)} = \frac{-i (n^2-1) \sin(k_1 d)}{2n \cos(k_1 d) - i (n^2+1) \sin(k_1 d)}$$

$$R = \left| \frac{E_{r10}}{E_{i10}} \right|^2 = \frac{(n^2-1)^2 \sin^2(k_1 d)}{4n^2 \cos^2(k_1 d) + (n^2+1)^2 \sin^2(k_1 d)} = \frac{(n^2-1)^2 \sin^2(k_1 d)}{4n^2 + (n+1)^2 (n-1)^2 \sin^2(k_1 d)}$$

$$R=0 \text{ 當且僅當 } k_1 d = l\pi \quad (l \text{ 為整數}), \text{ 即 } nd = \frac{l\pi}{k_1} = \frac{l\pi}{2\pi} \lambda = \frac{l c}{2f} = \frac{l \pi c}{\omega}$$